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Studierichting: Informatica
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Opgave: Bereken in $\mathbb{C}$

$$\frac{(1+i)^2}{(1-i)^3} + \frac{(1-i)^2}{(1+i)^3}$$

Oplossing:

$$\begin{aligned} & \frac{(1+i)^2}{(1-i)^3} + \frac{(1-i)^2}{(1+i)^3} \\ &= \frac{(1+i)^5}{(1-i)^3(1+i)^3} + \frac{(1-i)^5}{(1+i)^3(1-i)^3} \\ &= \frac{(1+i)^5 + (1-i)^5}{((1-i)(1+i))^3} \\ &= \frac{(2i)(2i)(1+i) + (-2i)(-2i)(1-i)}{(1+1)^3} \\ &= \frac{-4(i+1) - 4(i-1)}{8} \\ &= \frac{-4(1+1)}{8} \\ &= \frac{-4 \cdot 2}{8} \\ &= -1 \end{aligned}$$

References